

Chapter 10: Mechanics of Options Markets

Part 4 — Sections 10.10 to 10.12

Learning Objectives

After studying this part, you should be able to:

- Understand taxation issues related to options.
 - Understand the concept of constructive sale.
 - Understand warrants and how they differ from exchange-traded options.
 - Understand employee stock options and convertible bonds.
 - Understand the structure of over-the-counter (OTC) options markets.
 - Compare OTC and exchange-traded options.
 - Understand exotic options and customized derivatives.
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10.10 Taxation of Options

Tax treatment is an important consideration for option traders.

A profitable trading strategy can become unattractive if taxes significantly reduce returns.

Because of this, investors often consider:

- Timing of gains and losses
 - Tax rates
 - Tax deferral opportunities
 - Tax regulations regarding derivatives
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Tax Deferral Through Derivatives

Historically, investors sometimes used derivatives to:

- Lock in gains
- Delay tax payments

This was possible because ownership of the asset technically remained unchanged.

Example

Suppose an investor owns stock that has appreciated substantially.

The investor wishes to:

- Protect the gain
- Avoid immediate taxation

One possible approach is to establish a derivative position that offsets most of the risk.

Constructive Sale Rules

To prevent abuse, tax authorities introduced the concept of:

Constructive Sale

Constructive Sale Definition

A constructive sale occurs when:

An investor effectively disposes of an appreciated position without actually selling it.

The tax authority may then treat the position as if it were sold.

Situations Creating Constructive Sale

An appreciated asset may be considered constructively sold if the owner:

1. Enters a Short Sale

The investor shorts:

- The same asset
 - A substantially identical asset
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2. Enters a Futures or Forward Contract

The investor agrees to deliver:

- The same asset
- A substantially identical asset

in the future.

3. Eliminates Nearly All Risk and Reward

The investor enters one or more transactions that remove:

- Most downside risk
 - Most upside potential
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Why Constructive Sale Rules Exist

Without such rules:

Investors could:

- Lock in gains

- Avoid paying taxes indefinitely

Constructive sale rules ensure gains are recognized when economic ownership has effectively ended.

Important Exception

Reducing only downside risk generally does not trigger constructive sale treatment.

Example

Suppose an investor owns stock.

To protect against losses, the investor buys:

An in-the-money put option

This reduces downside risk.

However:

Upside potential still exists.

Therefore:

Constructive sale treatment usually does not apply.

Tax Planning and Options

Options can be used for:

Hedging

Reducing risk.

Tax Management

Managing timing of gains and losses.

Portfolio Restructuring

Changing exposures without immediately selling assets.

Caution

Hull emphasizes that:

Tax laws evolve continuously.

Before implementing tax-motivated option strategies:

Investors should carefully consider:

- Future legislative changes
 - Exit costs
 - Regulatory risks
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10.11 Warrants, Employee Stock Options, and Convertibles

Many option-like securities exist outside traditional exchange-traded options.

Important examples include:

1. Warrants
 2. Employee Stock Options
 3. Convertible Bonds
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Warrants

A warrant is:

An option issued by a corporation or financial institution.

Characteristics of Warrants

Unlike exchange-traded options:

The issuer creates the contract directly.

Examples:

- Gold warrants
 - Stock warrants
 - Currency warrants
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Corporate Warrants

Companies often issue warrants alongside bonds.

Purpose:

Make the bond issue more attractive.

Example

A company issues:

- Bond
- Attached call warrant

Investors receive:

- Fixed-income security

- Potential equity upside

This combination increases investor demand.

Key Feature of Warrants

When exercised:

The issuing company creates new shares.

This differs from exchange-traded options.

Exchange-Traded Call Option

When exercised:

Existing shares are transferred.

No new shares are created.

Warrant Exercise

When exercised:

The company issues new shares.

As a result:

Shares Outstanding Increase

Dilution Effect

Because new shares are created:

Existing shareholders may experience dilution.

Employee Stock Options (ESOs)

Employee stock options are:

Call options granted to employees by their employer.

Purpose of Employee Stock Options

Align employee incentives with shareholder interests.

Logic

If stock price rises:

Employees benefit.

Therefore employees are motivated to:

- Increase profitability
 - Improve performance
 - Increase firm value
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Typical Characteristics

Most ESOs are issued:

At-the-Money

Strike price approximately equals current stock price.

Accounting Treatment

Historically:

Many firms did not expense employee stock options.

Modern accounting standards generally require:

Employee stock options to be recognized as compensation expense.

Advantages of ESOs

Employee Motivation

Employees directly participate in firm success.

Retention

Employees are encouraged to remain with the company.

Cash Conservation

Compensation can be provided without immediate cash payments.

Convertible Bonds

A convertible bond is:

A bond that can be converted into company stock.

Structure

A convertible bond combines:

Straight Bond

Provides fixed-income payments.

PLUS

Call Option

Allows conversion into equity.

Embedded Option

The conversion feature effectively acts as:

A call option on the company's stock.

Why Investors Like Convertibles

They receive:

Downside Protection

Through bond characteristics.

Upside Potential

Through stock conversion rights.

Why Companies Issue Convertibles

Lower Interest Costs

Investors accept lower coupon rates.

Easier Financing

Conversion feature makes bonds attractive.

Comparing Warrants, ESOs, and Exchange-Traded Options

Feature	Warrants	ESOs	Exchange-Traded Options
Issuer	Company/Institution	Company	Market Participants
New Shares Created?	Yes	Yes	No
Dilution?	Yes	Yes	No
Standardized?	Usually No	No	Yes
Exchange Traded?	Sometimes	No	Yes

Important Hull Observation

For warrants and employee stock options:

Number of contracts is predetermined.

For exchange-traded options:

Number of contracts changes continuously as traders create or close positions.

10.12 Over-the-Counter Options Markets

Most of Chapter 10 focuses on exchange-traded options.

However:

The OTC market is enormous.

In many asset classes:

OTC trading exceeds exchange trading.

What is an OTC Option?

An OTC option is:

A privately negotiated option contract.

The contract is customized directly between counterparties.

Main Participants

OTC markets are dominated by:

Financial Institutions

Banks and dealers.

Corporate Treasurers

Managing financial risks.

Fund Managers

Hedging and portfolio management.

Popular OTC Underlyings

OTC options are especially common for:

Foreign Exchange

Currency options.

Interest Rates

Interest-rate options.

Credit Risk

Credit derivatives.

Major Advantage of OTC Markets

Customization.

Customized Features

Participants can negotiate:

Strike Price

Any desired strike.

Expiration Date

Any desired maturity.

Contract Size

Any desired notional amount.

Exercise Style

American, European, Bermudan, etc.

Example

A corporation may need:

- Maturity of 17 months
- Nonstandard strike price
- Unique notional size

Exchange-traded options may not provide these terms.

OTC contracts can be customized exactly.

Credit Risk in OTC Markets

Unlike exchange-traded options:

OTC contracts are not automatically guaranteed by a clearing organization.

Result

The option holder faces:

Counterparty Credit Risk

Counterparty Risk Example

Suppose:

- Option becomes highly profitable.
- Writer experiences financial distress.

The writer may fail to fulfill obligations.

This risk does not generally exist with OCC-cleared exchange options.

Collateral Requirements

To reduce credit risk:

Parties often post collateral.

Collateral provides protection if one side defaults.

Exotic Options

Many OTC options differ substantially from standard calls and puts.

These are known as:

Exotic Options

Definition

An exotic option is:

A nonstandard option with customized payoff characteristics.

Why Exotic Options Exist

Standard options may not perfectly hedge a firm's risk.

Exotic structures can be tailored to:

- Specific exposures
 - Unique market views
 - Complex hedging requirements
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Examples of Exotic Features

- Path-dependent payoffs
- Multiple exercise opportunities
- Average-price calculations
- Barrier triggers

These instruments are discussed extensively in later chapters.

Exchange-Traded vs OTC Options

Feature	Exchange-Traded	OTC
Standardized	Yes	No
Clearing House	Yes	Usually No
Credit Risk	Very Low	Significant
Liquidity	High	Depends on Dealer
Flexibility	Limited	Very High
Customization	Low	High

Advantages of Exchange-Traded Options

Transparency

Public pricing.

Liquidity

Active markets.

Reduced Credit Risk

Clearing house guarantees.

Ease of Trading

Standardized contracts.

Advantages of OTC Options

Customization

Contracts tailored exactly.

Flexibility

Any strike, maturity, or size.

Complex Structures

Supports exotic payoffs.

Disadvantages of OTC Options

Credit Risk

Counterparty may default.

Lower Transparency

Prices are privately negotiated.

Lower Liquidity

Position termination may require dealer negotiation.

Section Summary

Constructive Sale

Occurs when an investor effectively disposes of an appreciated position without formally selling it.

Warrants

Options issued by corporations or financial institutions.

Exercise creates new shares.

Employee Stock Options

Call options granted to employees.

Designed to align incentives.

Convertible Bonds

Bond + embedded call option.

Can be converted into equity.

OTC Options

Privately negotiated option contracts.

Major OTC Advantage

Customization.

Major OTC Disadvantage

Counterparty credit risk.

Exotic Options

Nonstandard options with customized payoffs.

Quick Revision Sheet

Constructive Sale

Effective disposal of an appreciated asset.

Warrant

Option issued by company or institution.

Warrant Exercise

Creates new shares.

Employee Stock Option

Call option granted to employees.

Convertible Bond

Bond + Call Option.

OTC Market

Privately negotiated derivatives market.

Main OTC Participants

- Banks
 - Corporate Treasurers
 - Fund Managers
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Main OTC Advantage

Customization.

Main OTC Risk

Counterparty default risk.

Exotic Option

Customized option with nonstandard payoff structure.

Exchange-Traded Options

Standardized and OCC-guaranteed.

OTC Options

Flexible but subject to credit risk.

End of Part 4 (Sections 10.10–10.12)

End of Chapter 10